# Hierarchical Music Analysis as Hypergraph Search

Introduction to Music Computing (Lecture 10)

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March 17, 2026

Durham University

# Overview

Hierarchical Music Analysis

Dynamic Programming

- Semiring Dynamic Programming
- Algebraic Dynamic Programming

Parsing as Hypergraph Search

- Non-Sequential Parsing
- Infinite and Continuous Alphabets

Monte Carlo Sampling

# Hierarchical Music Analysis

# A Generative Model of Musical Form [5]

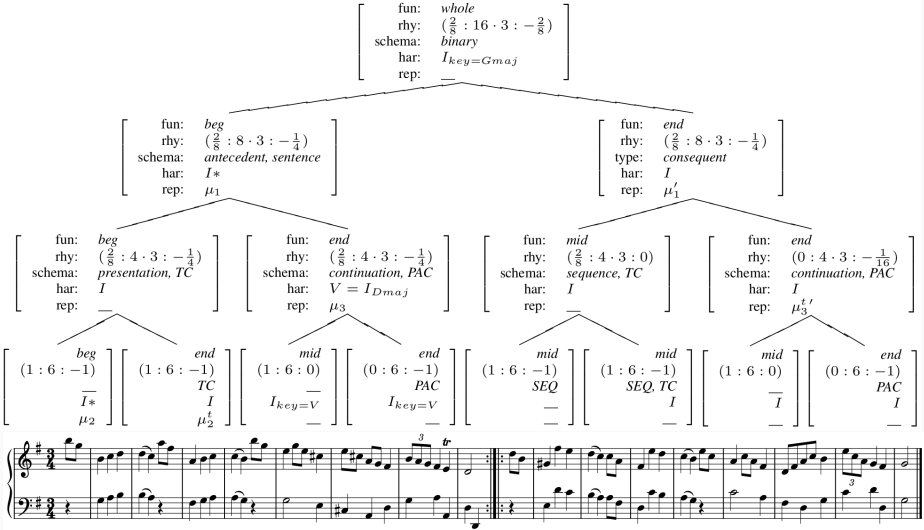
**Musical form is characterised by**

1. formal function
2. (hyper)metrical structure + rhythm grammar [3]
3. schemata and templates
4. harmonic anchors + harmonic grammar [1, 2, 4]
5. repetition structure

**Form categories $\mathcal{F}$**

$$c := \left[ \begin{array}{rl} \text{function}: & \in [beg, mid, end, whole] \\ \text{rhythm}: & [upbeat : body : coda], M \\ \text{schema}: & s \in \mathcal{S} \\ \text{harmony}: & chord \in \mathcal{H} \\ \text{repetition}: & v \in \mathcal{V} \end{array} \right] \in \mathcal{F}$$

# A Generative Model of Musical Form [5]




Mozart Minuet in G, K1

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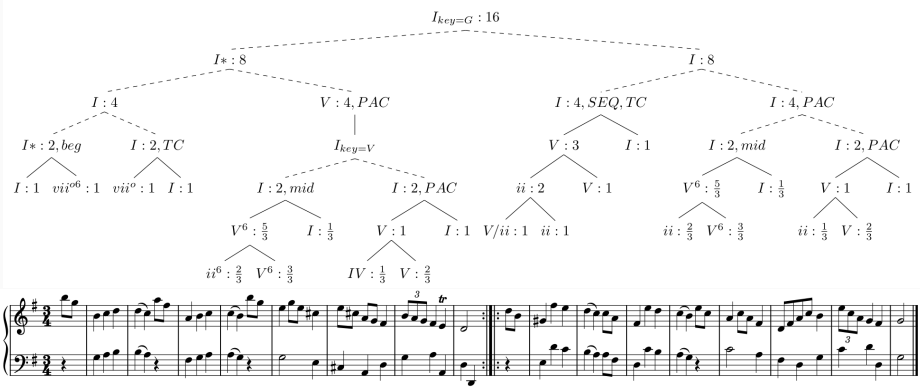


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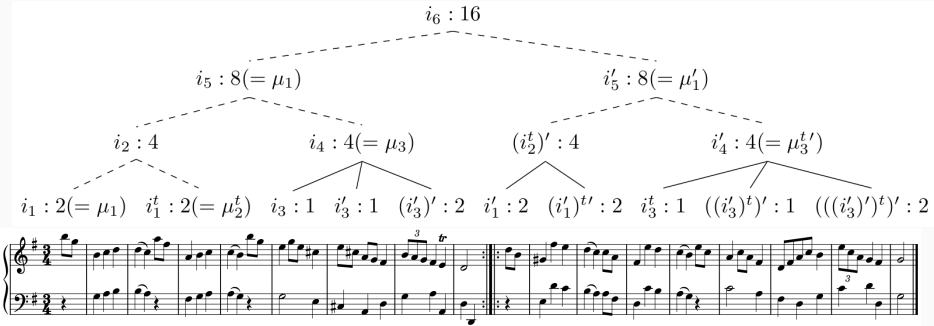


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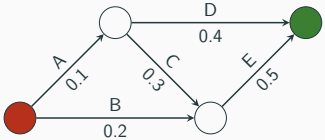
# Dynamic Programming

# Dynamic Programming (DP)

**Evaluate function $f$ over sets of elements $e$ from combinatorial space $\mathcal{C}$**




$\mathcal{C}$ : paths starting from red

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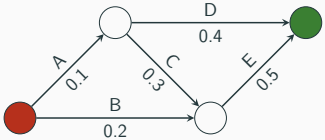


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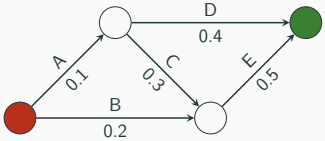


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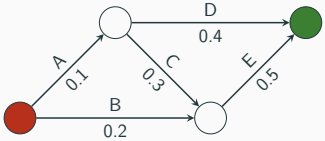
$$f(\{e_1, \ldots, e_n\}) = \bigoplus \left(f(e_1), \ldots, f(e_n)\right)$$

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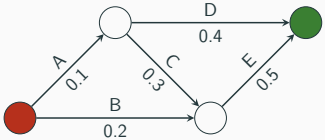


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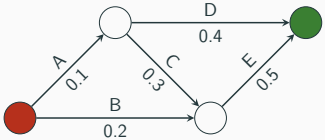


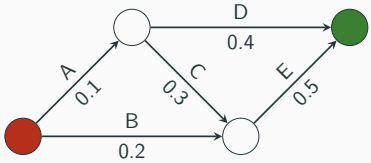

$\mathcal{C}$ : paths starting from red

$f$ : probability to reach green

$$\begin{aligned}
& f(< A, D >, < B, E >, < A, C, E >) \\
&= f(< A, D >) + f(< B, E >) + f(< A, C, E >) \\
&= f(A) \cdot f(D) + f(B) \cdot f(E) + f(A) \cdot f(C) \cdot f(E) \\
&= 0.1 \cdot 0.4 + 0.2 \cdot 0.5 + 0.1 \cdot 0.3 \cdot 0.5 \\
&= 0.155
\end{aligned}$$

# Dynamic Programming (DP)

**How to compute** $f(A) \cdot f(D) + f(B) \cdot f(E) + f(A) \cdot f(C) \cdot f(E)$ **efficiently?**



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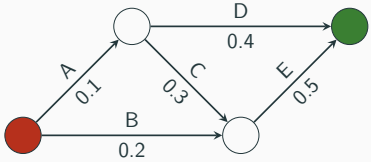
**How to compute** $f(A) \cdot f(D) + f(B) \cdot f(E) + f(A) \cdot f(C) \cdot f(E)$ **efficiently?**

**Factorise:** $f(A) \cdot f(D) + \big(f(B) + f(A) \cdot f(C)\big) \cdot f(E)$



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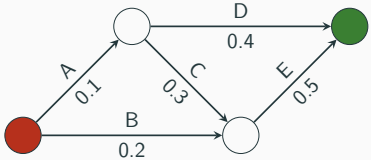
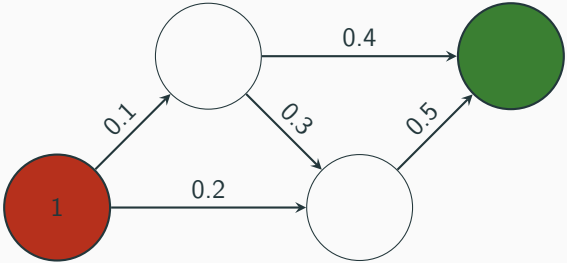
$\underbrace{\phantom{x}}_{x_5}$




$$
\begin{aligned}
& x_1 = f(A) \cdot f(D) \\
& x_2 = f(A) \cdot f(C) \\
\longrightarrow \quad & x_3 = f(B) + x_2 && = f(B) + f(A) \cdot f(C) \\
& x_4 = x_3 \cdot f(E) && = f(B) \cdot f(E) + f(A) \cdot f(C) \cdot f(E) \\
& x_5 = x_1 + x_4 && = f(A) \cdot f(D) + f(B) \cdot f(E) + f(A) \cdot f(C) \cdot f(E)
\end{aligned}
$$

# Marginal Likelihood (Sum/Prod)



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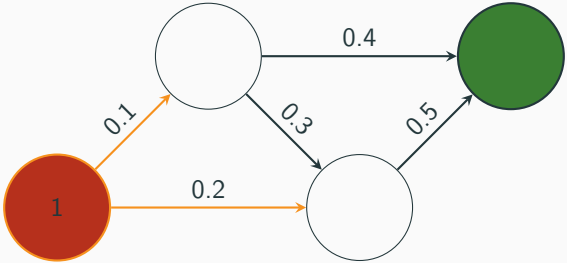

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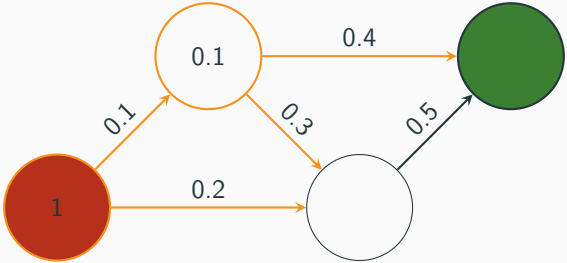

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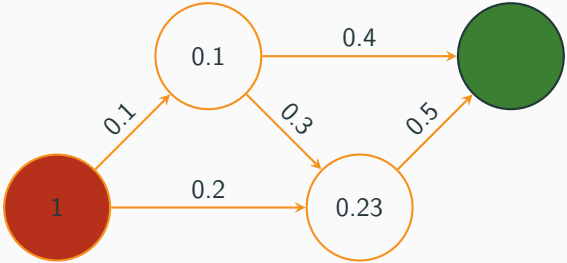

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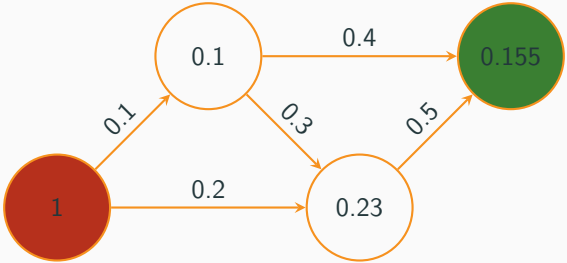


# Dynamic Programming

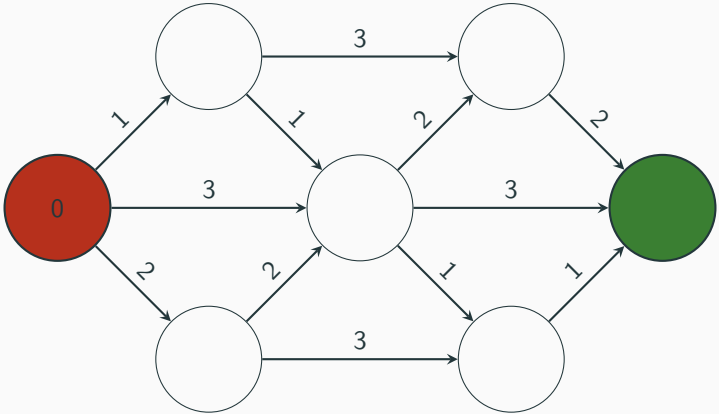
## Semiring Dynamic Programming

# Semiring Dynamic Programming

**We can define $\oplus$ and $\otimes$ how ever we want!**

| **Semiring** | $\oplus$ | $\otimes$ | **Values** |
|---|---|---|---|
| **Sum/Prod** | $+$ | $\cdot$ | marginal probabilities |
| **Bool** | $\vee$ (`or`) | $\wedge$ (`and`) | correctness/existence |
| **LSE/Sum** | $\log \sum \exp$ | $+$ | marginal log-probabilities |
| **Min/Sum** | min | $+$ | minimal costs / shortest path |
| **Max/Prod** | max | $\cdot$ | highest probability (Viterbi) |
| **Count** | $+$ | $\cdot$ | counter/integer |
| $k$-**Max/Prod** | $k$-max | $\otimes$ (outer product) | $k$-best |

# Shortest Path (MinSum)



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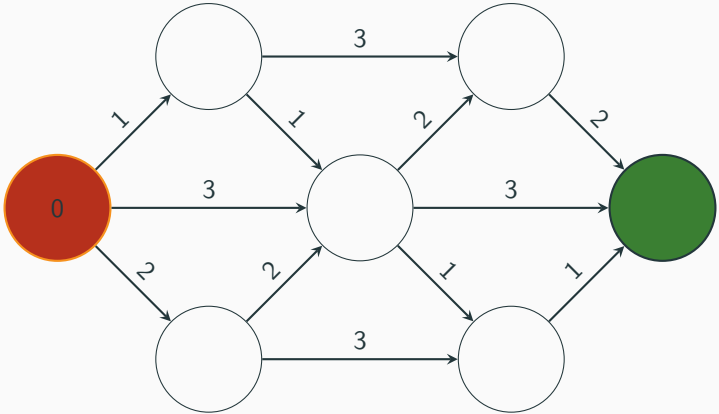

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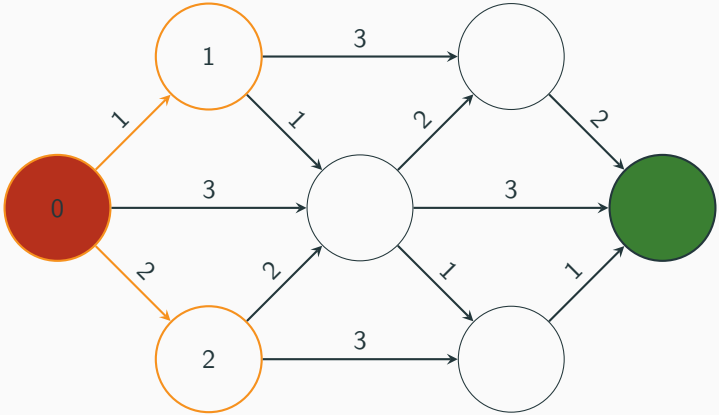

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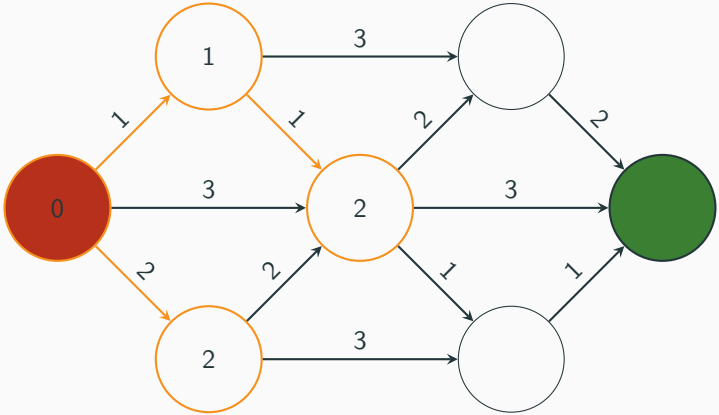

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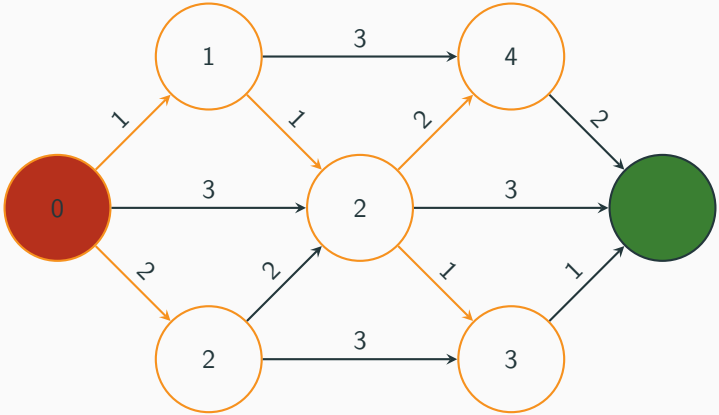

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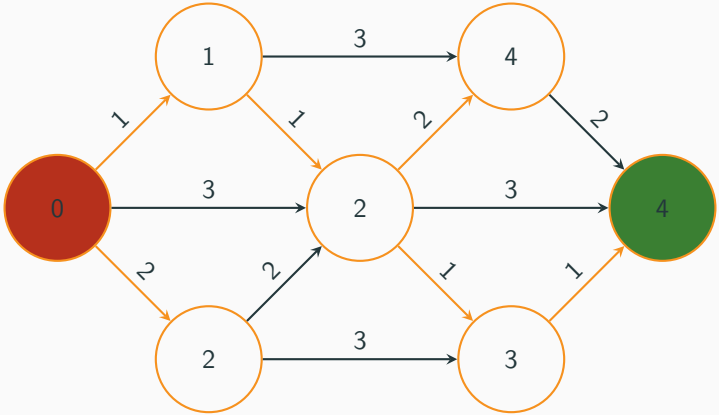

# Dynamic Programming

## Algebraic Dynamic Programming

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**Signature** for combining elements

$$\{\mathcal{C}^m \to \mathcal{C}^n \mid \text{usually } m > 1 \text{ and } n = 1\}$$

**Grammar** rules $r \in \mathcal{R}_{\mathcal{C}}$ according to signature

$$r : \mathcal{C}^m \to \mathcal{C}^n$$

**Algebra** for value updates according to signature

$$\bigotimes : \mathcal{V}^m \to \mathcal{V}^n$$

**Choice function** / aggregation operation

$$\bigoplus : \mathcal{V}^* \to \mathcal{V}$$

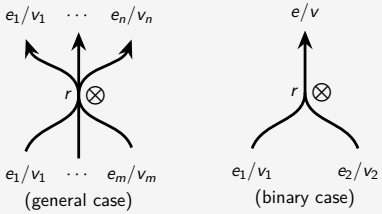
## Paths in a Graph

**Grammar:**

$r_1 : \text{edge}(\mathit{start}, Y) \to \text{path}(\mathit{start}, Y)$
$r_2 : (\text{path}(\mathit{start}, X), \text{edge}(X, Y)) \to \text{path}(\mathit{start}, Y)$

**Algebra** (multiplication): $\bigotimes$
**Aggregation** (addition): $\bigoplus$

## Signature as Hyperedges



# Marginal Likelihood [revisited]

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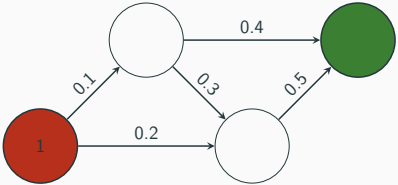

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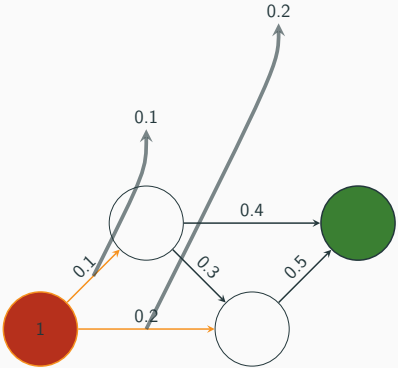

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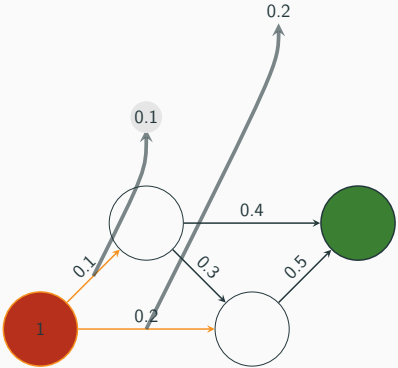

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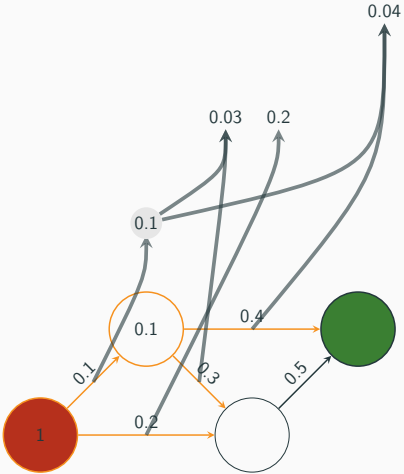

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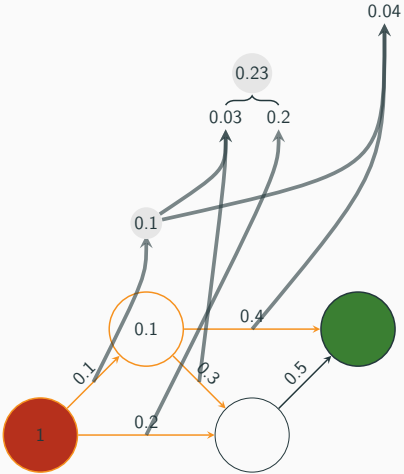

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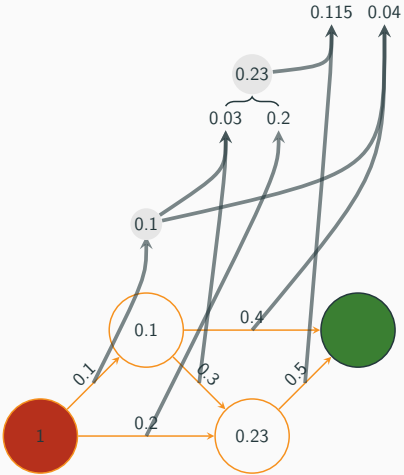

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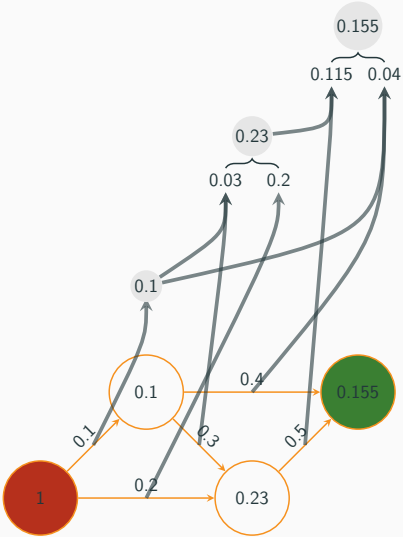

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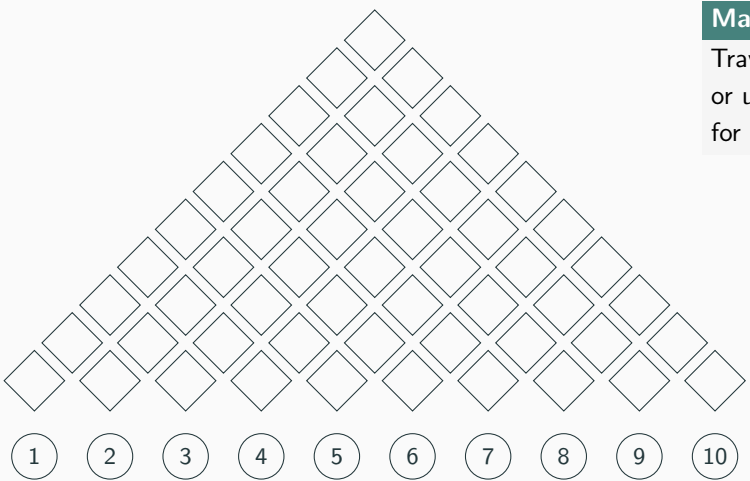


**Marginalisation**

Traverse complete hypergraph or use Monte Carlo sampling for approximation.

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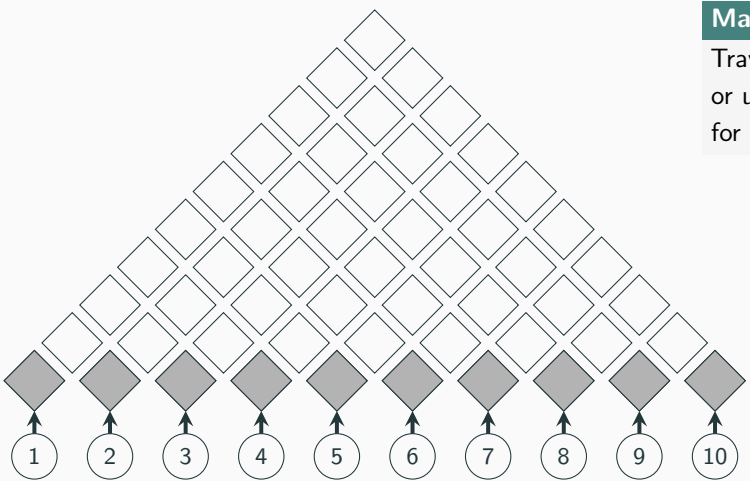


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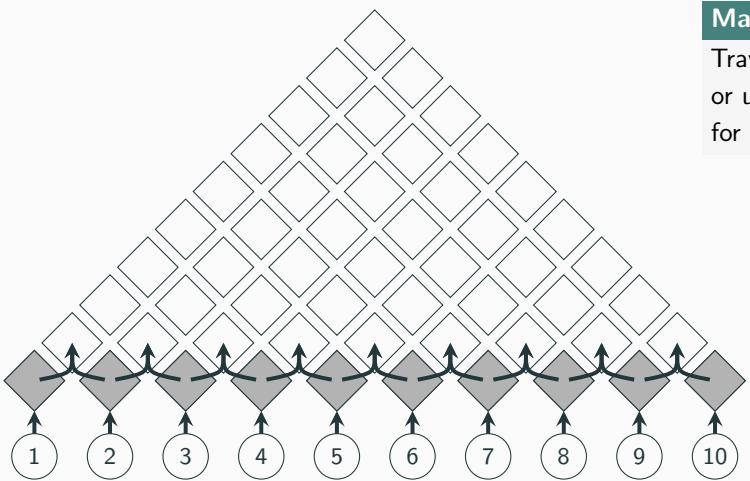
# Parsing as Hypergraph Search

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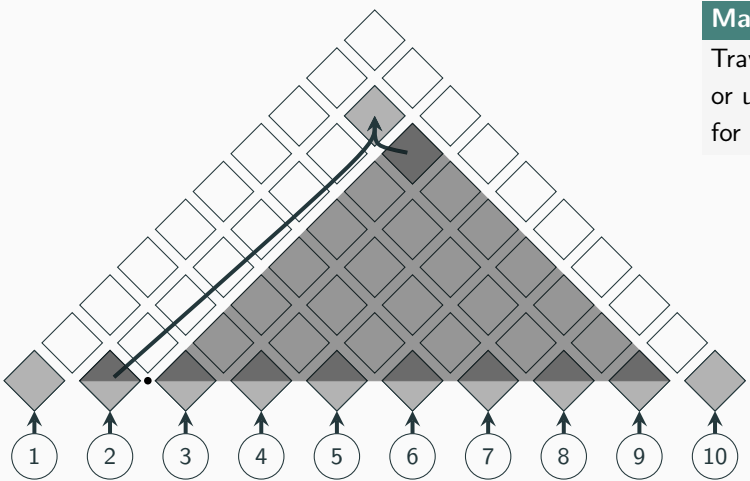


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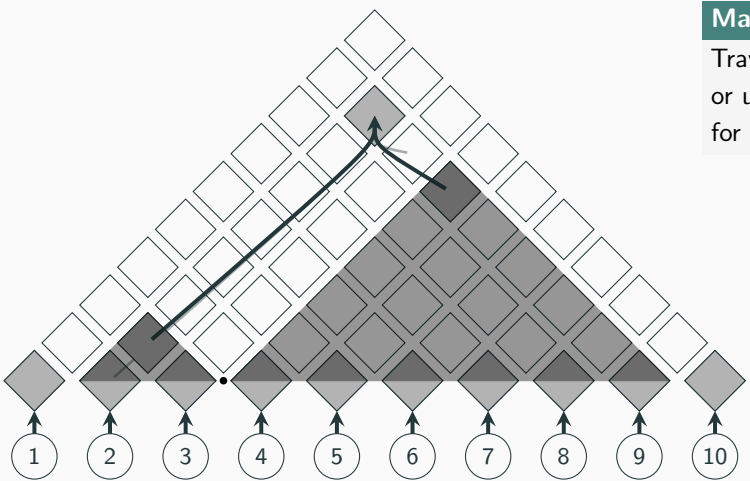


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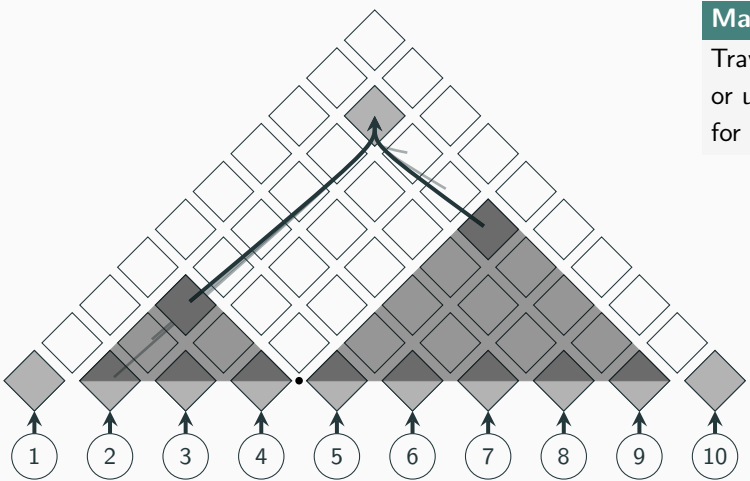


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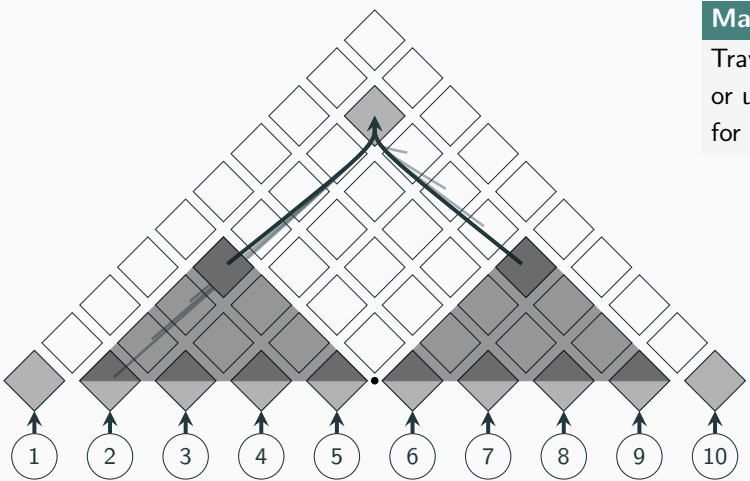


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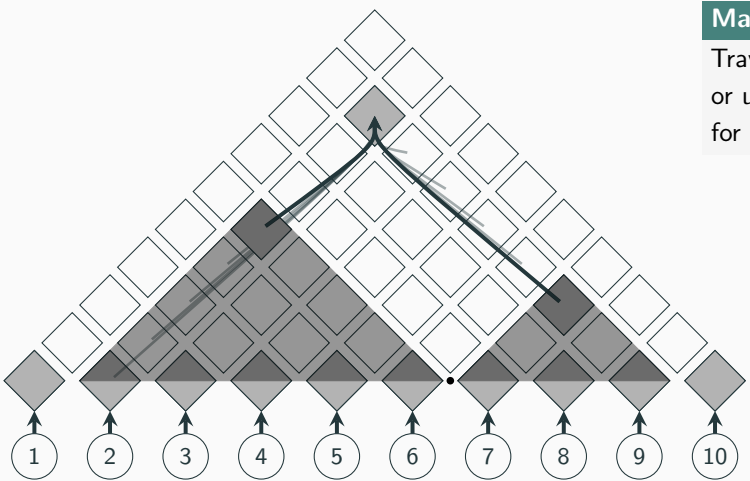


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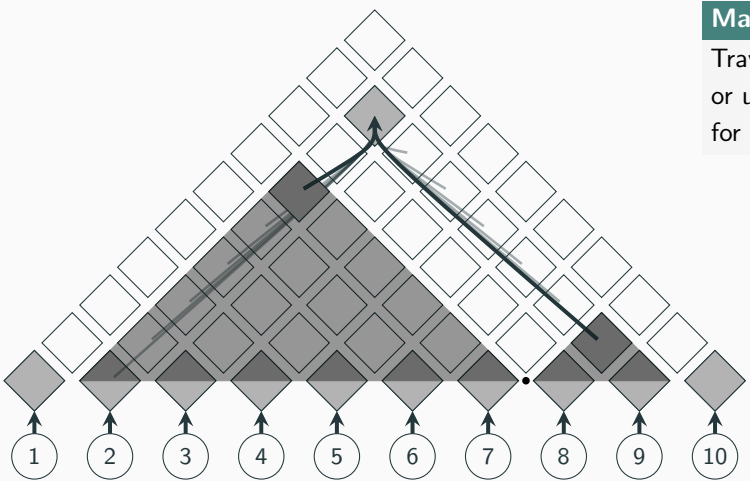


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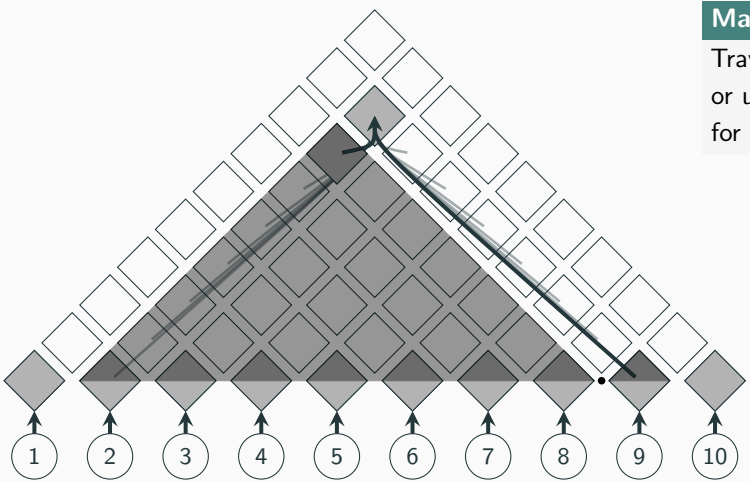


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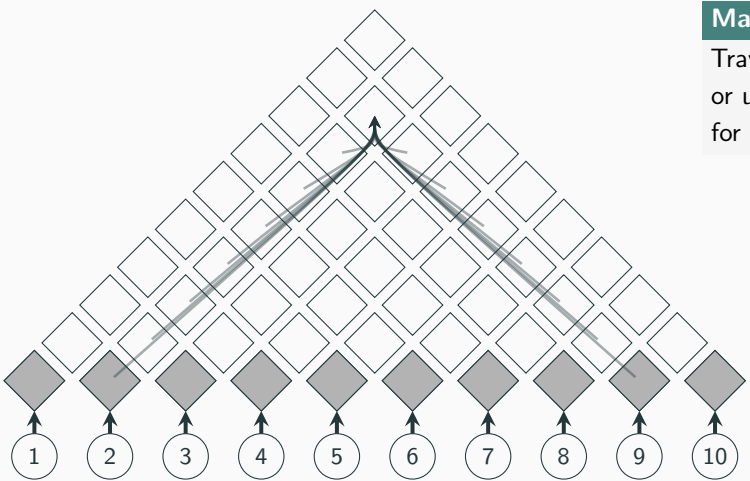


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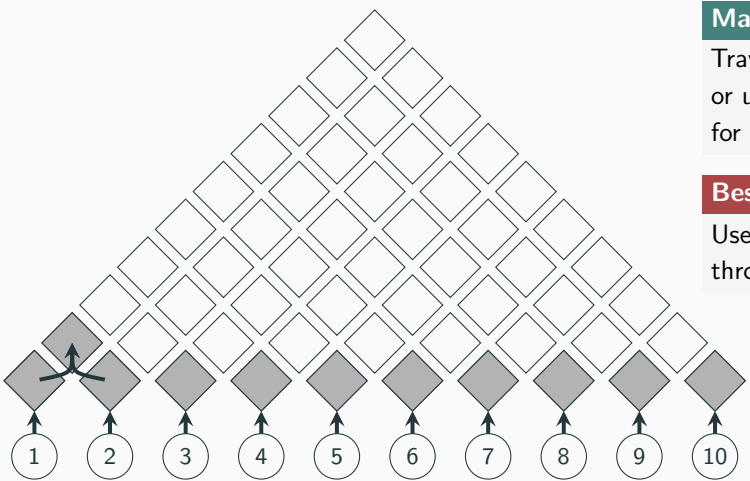


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**Best Parse Tree**

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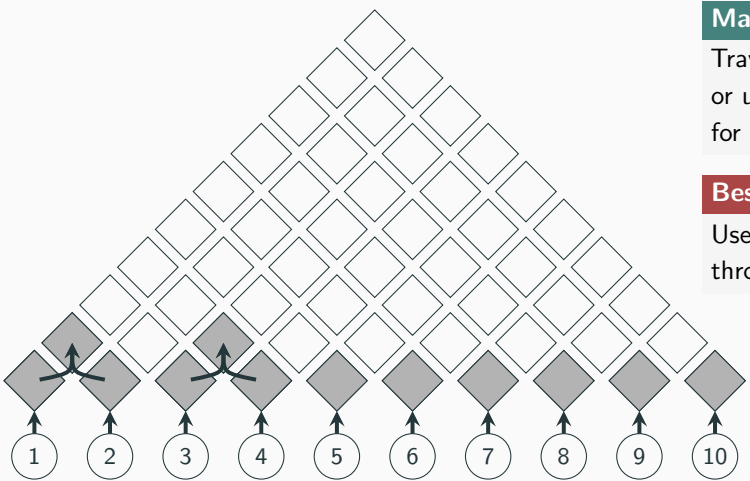


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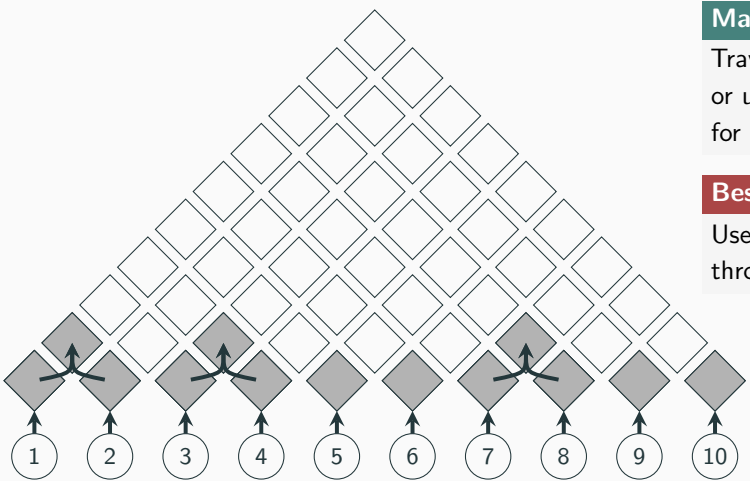


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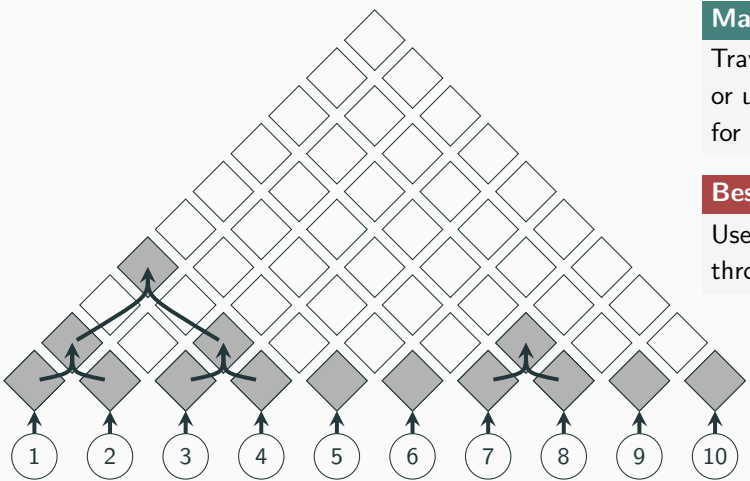


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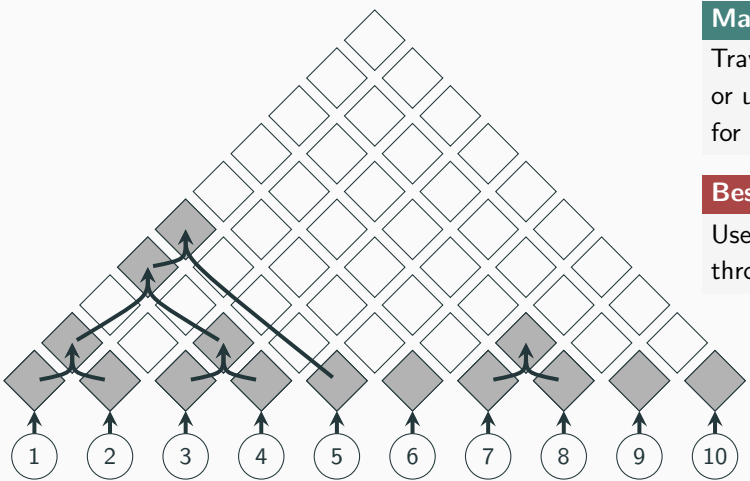


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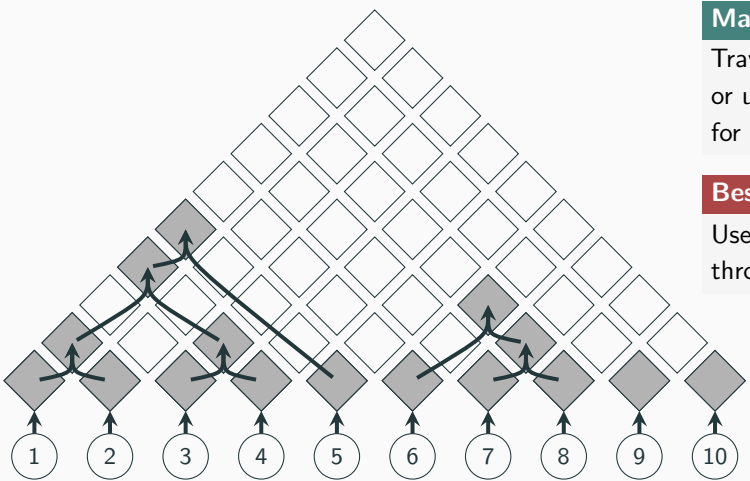


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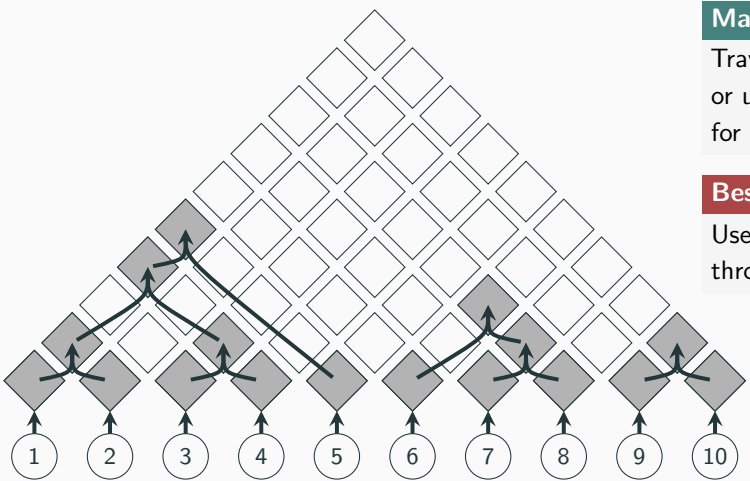


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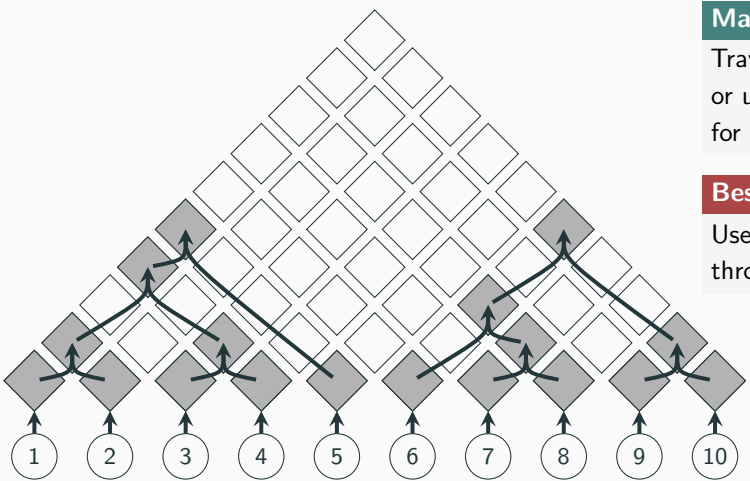


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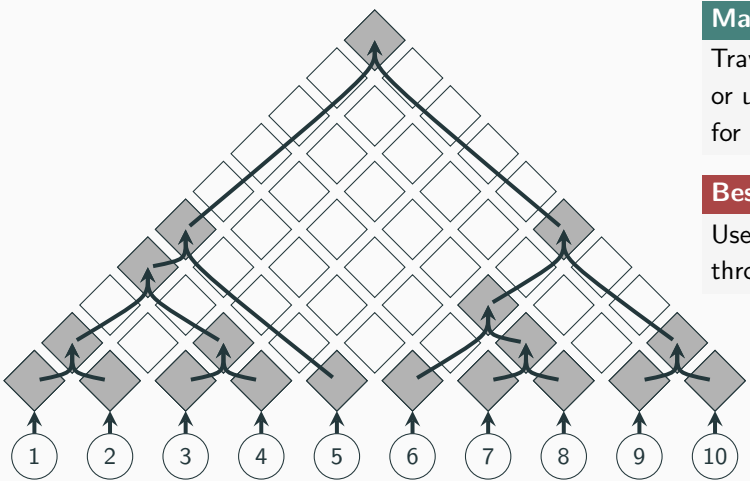


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# Parsing as Hypergraph Search

## Non-Sequential Parsing

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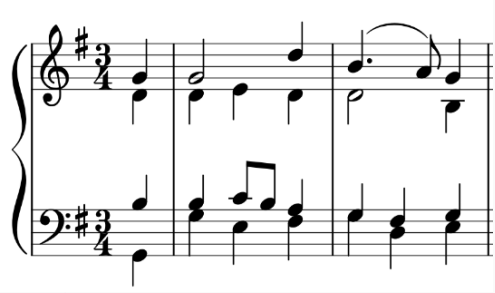

J.S.Bach Aus meines Herzens Grunde

## Rules

$r_1$ : note $\rightarrow$ voice

$r_2$ : (voice, follow-up-note) $\rightarrow$ voice

$r_3$ : note $\rightarrow$ (note, . . . , note)

$r_4$ : (voice, . . . , voice) $\rightarrow$ piece

## Algebra

$r_1$ : high fixed cost

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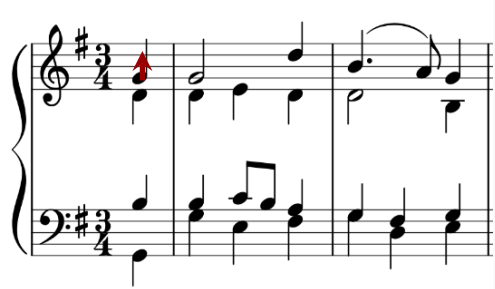

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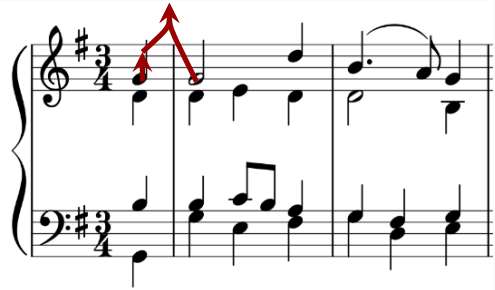

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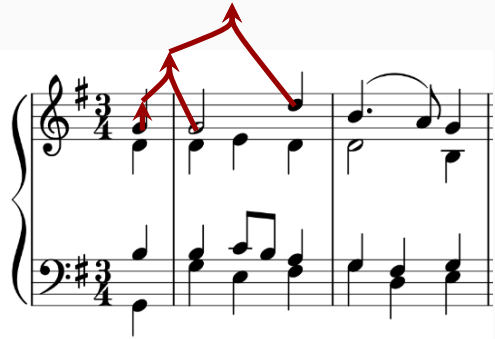

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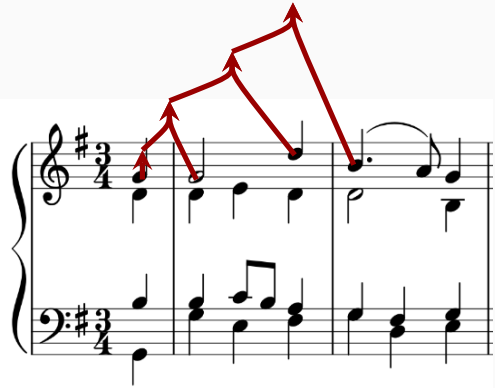

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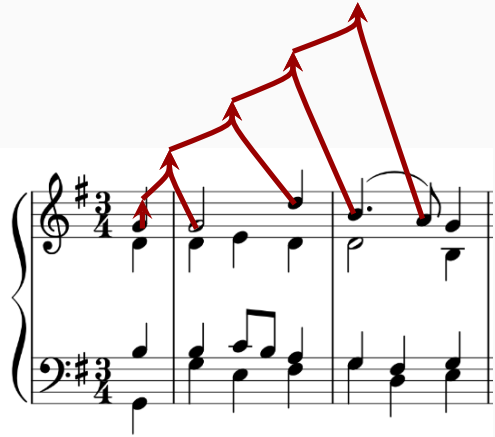

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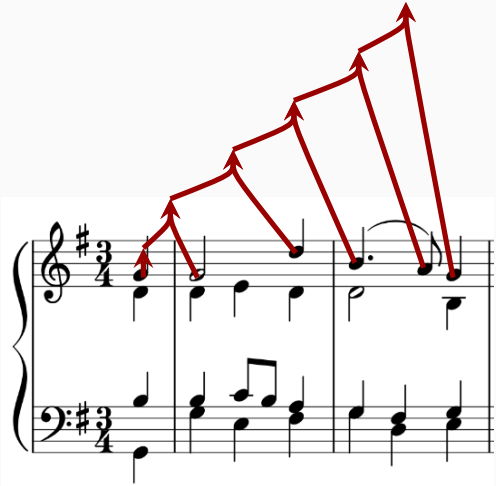

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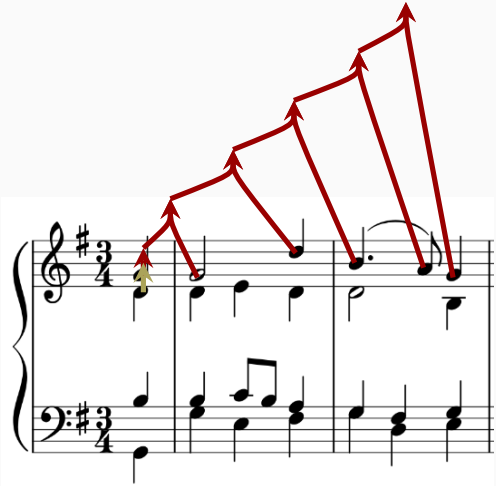

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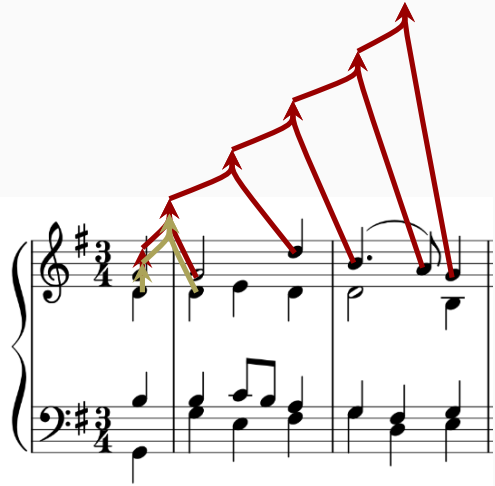

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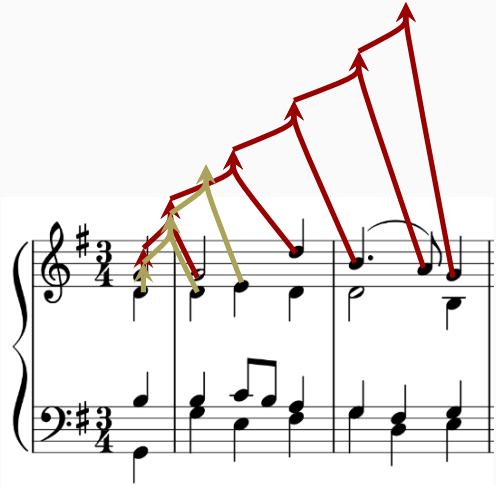

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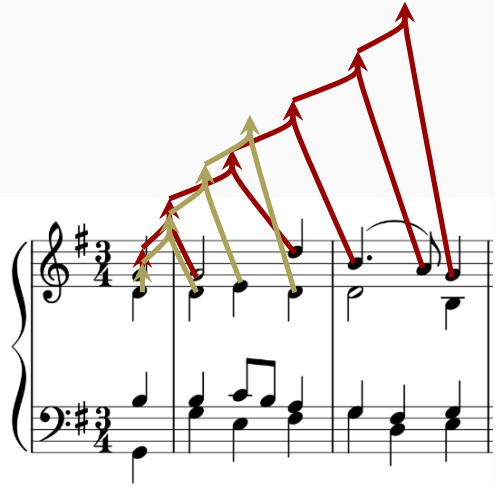

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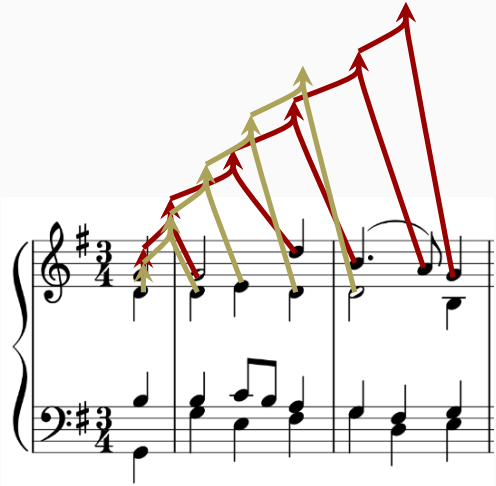

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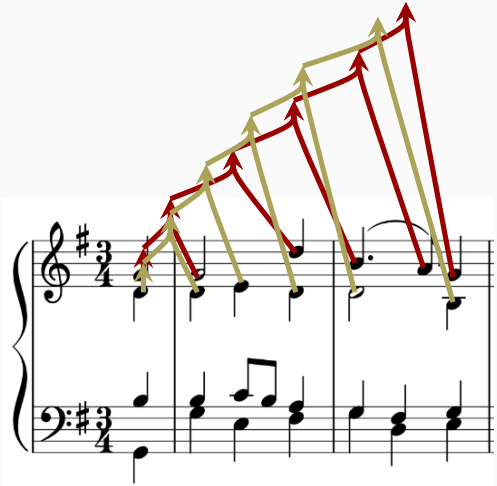

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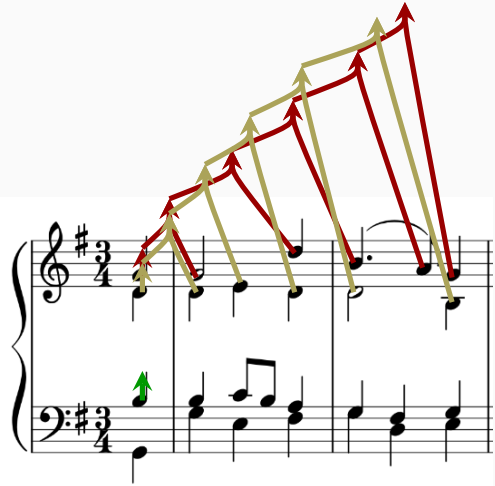

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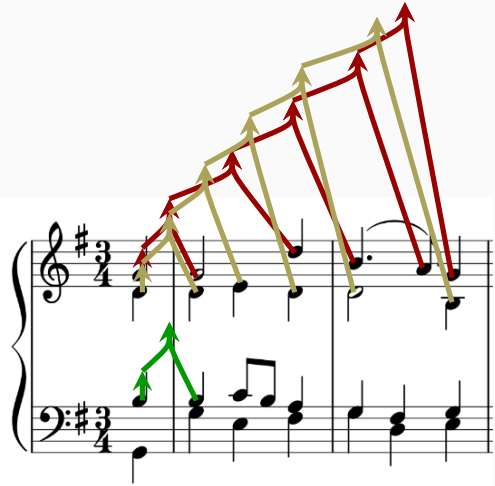

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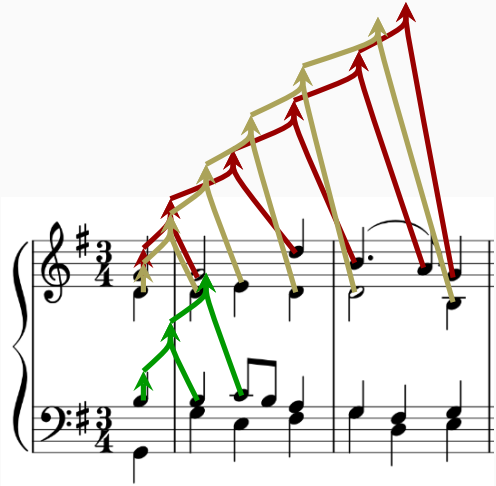

J.S.Bach Aus meines Herzens Grunde

## Rules

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## Algebra

$r_1$ : high fixed cost

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$\bigoplus$ : min

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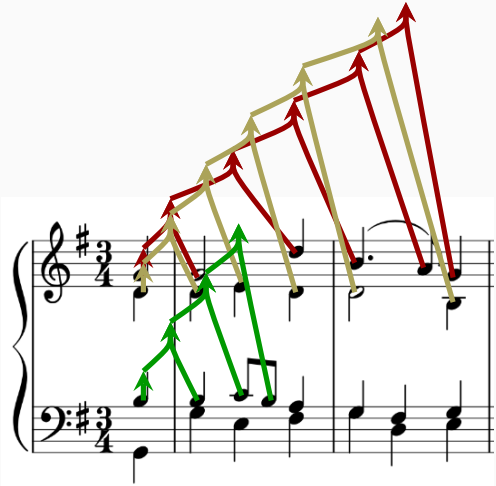

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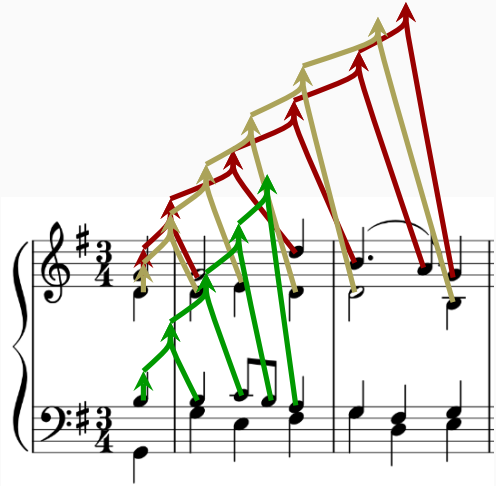

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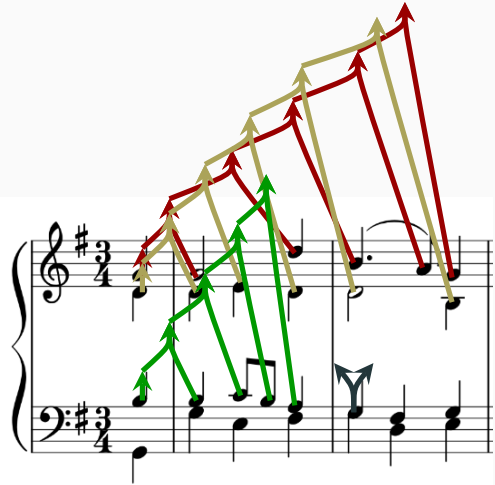

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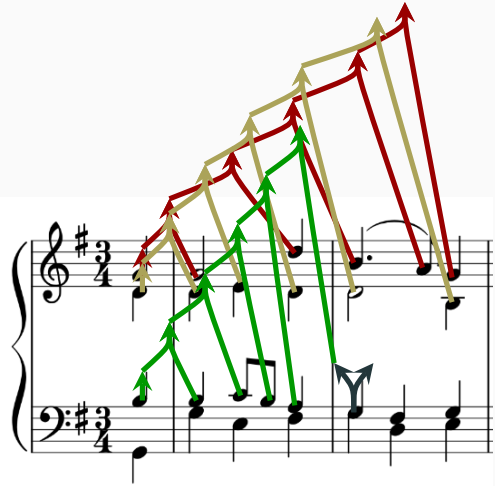

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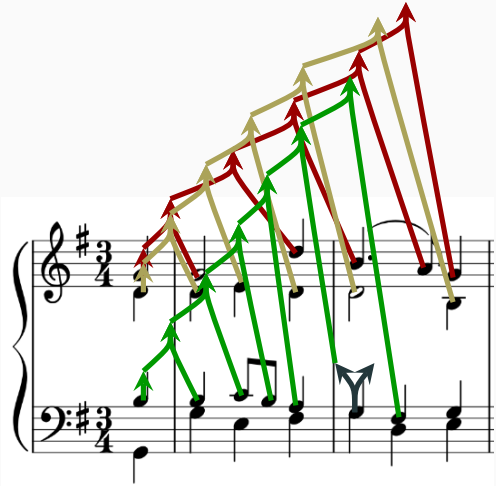

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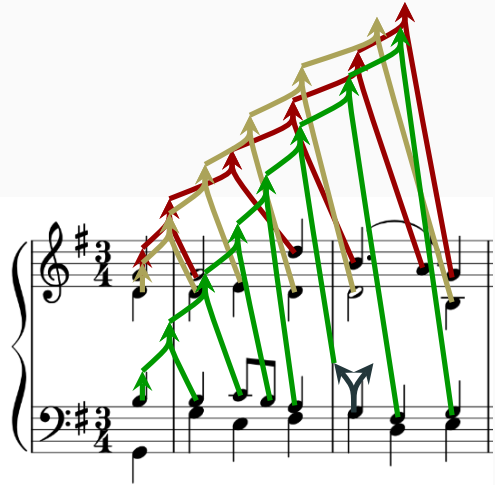

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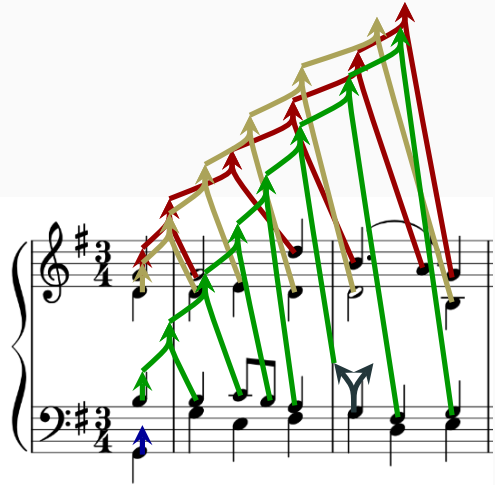

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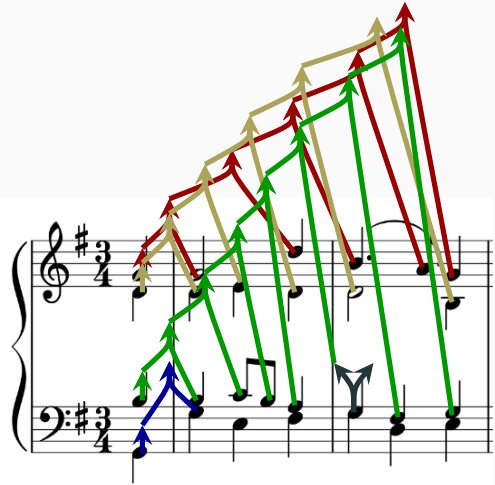

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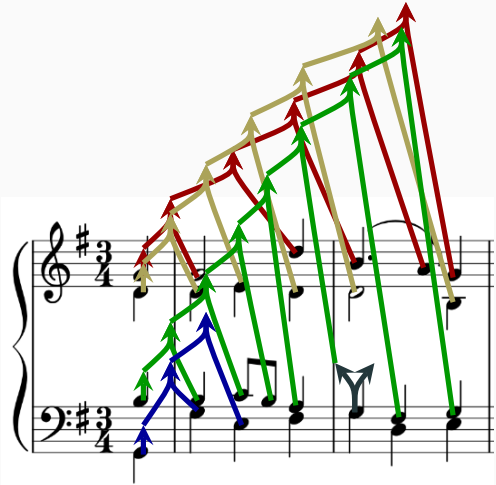

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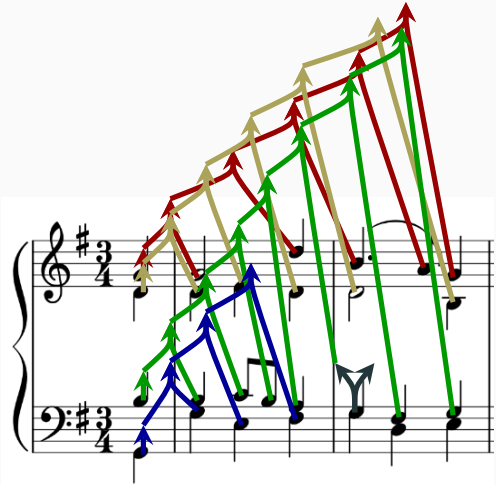

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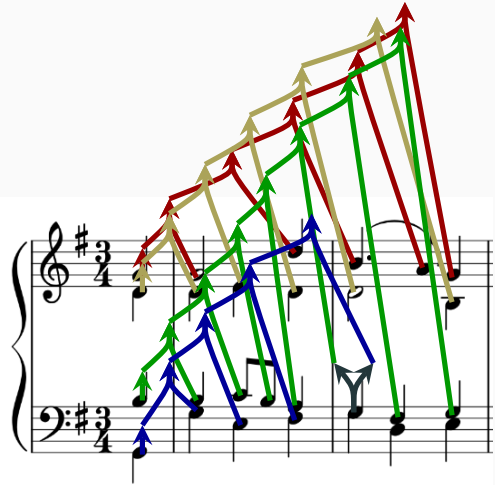

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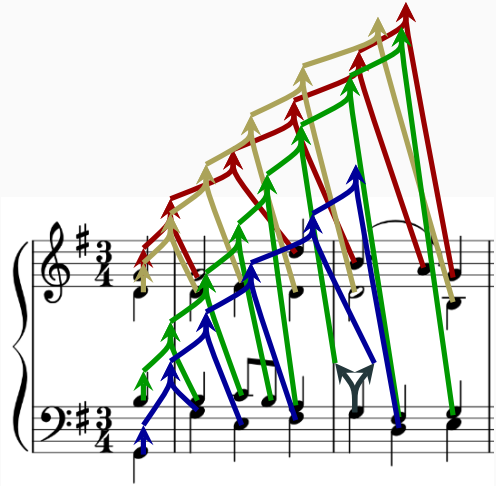

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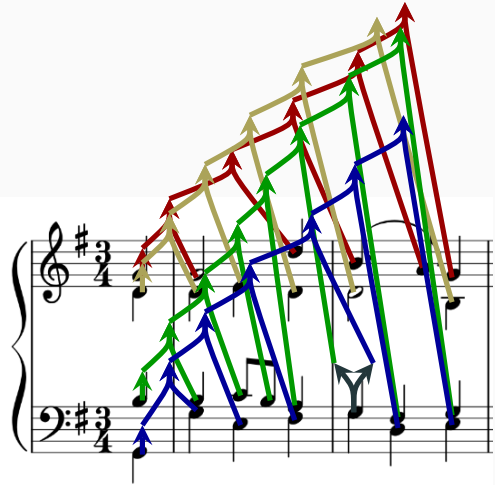

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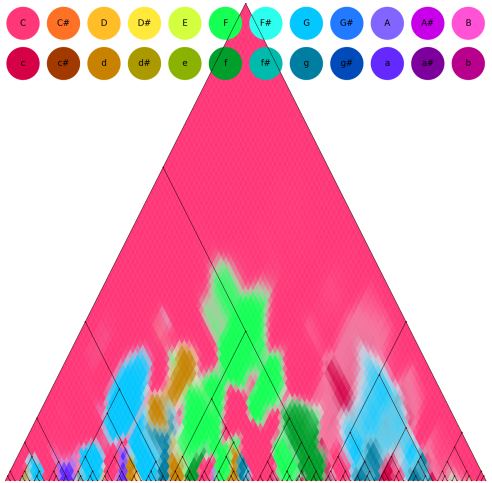
# Parsing as Hypergraph Search

## Infinite and Continuous Alphabets

# Pitch-Class Distribution (PCD) Parsing




## Rules

$$\mathcal{R} = \left\{(\mathrm{PCD}^{\tau}_{i:j}, \mathrm{PCD}_{j:k}) \to \mathrm{PCD}_{i:k}\right\}$$

for $\quad i < j < k \in \mathbb{N}$

and $\quad \tau$ : any transposition

## Algebra

$$p(\mathrm{PCD}_{i:k}) = p(\mathrm{PCD}^{\tau}_{i:j}) \cdot \mathrm{Dir}(\mathrm{PCD}^{\tau}_{i:j} \mid \mathrm{PCD}_{i:k}) \cdot p(\mathrm{PCD}_{j:k}) \cdot \mathrm{Dir}(\mathrm{PCD}_{j:k} \mid \mathrm{PCD}_{i:k}) \cdot \mathrm{Beta}\left(\frac{j-i}{k-i}; \alpha, \beta\right) \cdot p(\tau)$$

## Choice/Aggregation

$\bigoplus$ : max

# Monte Carlo Sampling

# Monte Carlo Sampling

## Importance Sampling

**Basic Idea**

- Sample tree $t$ from the hypergraph with *proposal* probability $q(t)$
- Compute *actual* probability $p(t)$
- Compute importance sampling weight $w(t) = p(t)/q(t)$
- Approximate any expectations $\mathbb{E}_{p(t)}[f(t)] \approx \frac{1}{N}\sum_{i=1}^{N} f(t_i)\, w(t_i)$
- Approximate marginal likelihood $p(t) \approx \frac{1}{N}\sum_{i=1}^{N} w(t_i)$
- Train parameters by maximising $p(t)$

## Sequential Monte Carlo

**Problem**

- Many samples run into dead ends
- Resample particles according to $w(t)$
- Reset $w(t) = 1$ but track $Z_\tau = \frac{1}{N}\sum_{i=1}^{N} w(t_i)$
- Approximate any expectations as before
- Approximate marginal likelihood $p(t) \approx \prod_{\tau=1}^{T} Z_\tau$

## Self-Normalised Importance Sampling

- For unnormalised $p(t) = \tilde{p}(t)/Z$ self-normalise weights $\bar{w}(t) = w(t)/\sum_i w(t_i)$
- Cannot estimate $p(t)$

# References


[1] Martin Rohrmeier.
**Towards a generative syntax of tonal harmony.**
*Journal of Mathematics and Music*, 5(1):35–53, March 2011.

[2] Martin Rohrmeier.
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[3] Martin Rohrmeier.
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[4] Martin Rohrmeier and Fabian C. Moss.
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